PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Approaches and outcomes of community health worker's interventions for hypertension management and control in low- and middle-income countries: Systematic review
AUTHORS	Mbuthia, Grace; Magutah, Karani; Pellowski, Jennifer

VERSION 1 – REVIEW

REVIEWER	Yan, Lijing Global Health Research Center, Duke Kunshan University, Kunshan, Jiangsu, China
REVIEW RETURNED	16-Aug-2021

GENERAL COMMENTS	This review was conducted by both Enying Gong (ORCID: 0000-0003-2203-0369) and Lijing L. Yan (0000-0002-3029-0462). In this systematic review, the authors reviewed existing trials that evaluated community health worker-led interventions for blood pressure intervention and control. The study design followed the basic systematic review requirement with a clear definition of PICOs in methods, but there are some major issues: 1. Abstract: The abstract has not covered some key items based on the PRISMA guidelines. The following items were missing: the methods used to assess the risk of bias, the methods used to synthesize results, and the brief summary of the limitation of the evidence included in the review. The conclusion is a little overstated from the results as the result section did not emphasize the integration between CHW-led services and primary healthcare. 2. Background: In the background, the author mentioned the huge issue of blood pressure control in LMICs, the definition of CHWs. The author stated that the previous reviews focused on NCD but not hypertension, which is not true. There are a few existing review articles that focused on the role of CHWs, training programs, and the task shifting and sharing strategies in BP management. The research gap has not been proposed upon the most recent literature in this field. o Anand, T. N., Joseph, L. M., Geetha, A. V., Prabhakaran, D., & Jeemon, P. (2019). Task sharing with non-physician health-care workers for management of blood pressure in low-income and middle-income countries: a systematic review and meta-analysis. The Lancet Global Health, 7(6), e761-e771. o Mills, K. T., Obst, K. M., Shen, W., Molina, S., Zhang, H. J., He, H., ... & He, J. (2018). Comparative effectiveness of implementation strategies for blood pressure control in hypertensive patients: a systematic review and meta-analysis. Annals of internal medicine, 168(2), 110-120. o Abdel-All, M., Putica, B., Praveen, D., Abimbola, S., & Joshi, R. (2017). Effectiveness of community health worker training programmes for cardiovascular disease management in

	low-income and middle-income countries: a systematic review. BMJ open, 7(11), e015529. 3. The objectives of this review are clearly stated with two objectives: what were the main types of CHW-based interventions, and what were the impacts of CHW-led interventions. However, the research findings cannot adequately answer the research question regarding the impacts of CHW-led intervention. 4. In the methods section, the author should clearly state how they classified intervention types and outcomes in the data extraction stage and how they synthesized data and information across studies. 5. Search strategy: o Alternative terms beyond CHW or volunteers, or lay workers can be added such as allied health professionals. o Use of developing countries or LMICs in the search strategy is problematic as it may miss certain studies even if used as MESH. o Similarly, only use RCT or N-R CT in MESH to identify study types may miss certain studies. 6. The result section has clearly described the characteristics of each study. However, the evidence synthesis has been done on a very superficial level. CHW-involved interventions could include several key types by considering the provider and receptor of the intervention, the dosage and contents. Many of the included studies have multiple components, and CHWs could be considered as both intervention providers and receptors. However, the tables and texts remain descriptive and the key messages learned were not distilled to a higher level. 7. Similarly, the result section of the outcome and main finding section and Table 1 described the key findings from the trial in a very superficial way. It only described key outcomes in trials, but not the trends in study findings and how the impact of the intervention could be different across studies. We recognize a meta analysis may not be possible; but synthesizing beyond description of individual studies will be more helpful and expected from the review. 8. Discussion section: The discussion is generic to the importance of CHW in the healthcare system and the feasibility and positive impact. There is an increasing number of studies in this field, and the general feasibility and impact have been well-known. The authors haven't discussed how the study findings from this review added to previous reviews or studies in the field and how the study findings could shed light for future intervention design. 9. There are some typos/language and formatting issues that need to be improved throughout the manuscripts. 10. The study followed PRISMA guideline 2009. There is PRISMA 2020, and we suggest using the new guideline.

REVIEWER	Perry, HB John Hopkins, International Health
REVIEW RETURNED	16-Nov-2021

GENERAL COMMENTS	Congratulations on a high-quality paper about a topic of great global public health importance. I recommend acceptance of the paper as is with some minor provisos.

	First, there are scattered grammatical and typographical errors throughout the paper. I presume the proofing editors will make sure these are all corrected. Secondly, I question whether a pre-test/post-test design qualifies as a quasi-experimental method. My understanding is that quasi-experimental designs consist of studies with non-randomized control groups. With the pre-test/post-test design, there is no control group. I did a brief search and did find some literature that considered the pre-test/post-test design as a quasi-experimental design, but I suspect that the weight of expert opinion is that this is not a quasi-experimental design. The gold standard regarding this issue is still, I presume, the classic book Experimental and Quasi-Experimental Designs for Research by Donald T. Campbell and Julian C. Stanley. The authors should consult this book or some other authoritative text to just their assertion the pre-test/post-test desires are quasi-experimental. Nonetheless, I do not have any objections to including the two articles with the pre-test/post-test design in the review. Thirdly, on line 48 of page 3 the authors cite 4 references supporting the statement that CHWs are increasingly recognized as an essential part of the health workforce needed to achieve public health goals. Another commonly cited reference to justify this statement is the following reference: Perry H, Zulliger R, Rogers, M. Community health workers in low-, middle- and high-income countries: an overview of their history, recent evolution, and current effectiveness. Annual Review of Public Health 2014;35:399-421. (Confession: I am one of the authors of this paper. Nonetheless, it is a widely cited paper.) Fourthly, on p. 5, line 22, the text states that the studies included had a comparison area. This is not correct, since the two studies with a pre-test/post-test design did not have a comparison area.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: First, there are scattered grammatical and typographical errors throughout the paper. I presume the proofing editors will make sure these are all corrected.
Authors: We have corrected the grammatical and typographical errors.

Reviewer 1: Secondly, I question whether a pre-test/post-test design qualifies as a quasi-experimental method. My understanding is that quasi-experimental designs consist of studies with non-randomized control groups. With the pre-test/post-test design, there is no control group. I did a brief search and did find some literature that considered the pre-test/post-test design as a quasi-experimental design, but I suspect that the weight of expert opinion is that this is not a quasi-experimental design. The gold standard regarding this issue is still, I presume, the classic book Experimental and Quasi-Experimental Designs for Research by Donald T. Campbell and Julian C. Stanley. The authors should consult this book or some other authoritative text to just their assertion the pre-test/post-test desires are quasi-experimental. Nonetheless, I do not have any objections to including the two articles with the pre-test/post-test design in the review.

Authors: Thank you for this enlightening comment and the informative resource you have given we have clarified in the inclusion criteria that in addition to RCTs and quasi experimental studies we also included the pretest and posttest studies.

Reviewer 1: Thirdly, on line 48 of page 3 the authors cite 4 references supporting the statement that CHWs are increasingly recognized as an essential part of the health workforce needed to achieve public health goals. Another commonly cited reference to justify this statement is the following reference: Perry

H, Zulliger R, Rogers, M. Community health workers in low-, middle- and high-income countries: an overview of their history, recent evolution, and current effectiveness. Annual Review of Public Health 2014;35:399-421. (Confession: I am one of the authors of this paper. Nonetheless, it is a widely cited paper.)

Authors: Thank you for the suggested informative article, we regret missing it in the first place! we have now included it as part of our reference.

Reviewer 1: Fourthly, on p. 5, line 22, the text states that the studies included had a comparison area. This is not correct, since the two studies with a pre-test/post-test design did not have a comparison area.

Authors: We have revised the inclusion to indicate that pretest and posttest studies were also included.

Reviewer 2: In this systematic review, the authors reviewed existing trials that evaluated community health worker-led interventions for blood pressure intervention and control. The study design followed the basic systematic review requirement with a clear definition of PICOs in methods, but there are some major issues:
The abstract has not covered some key items based on the PRISMA guidelines. The following items were missing: the methods used to assess the risk of bias, the methods used to synthesize results, and the brief summary of the limitation of the evidence included in the review. The conclusion is a little overstated from the results as the result section did not emphasize the integration between CHW-led services and primary healthcare.
Authors: Thank you for your observation, we have now included the method used to asses the risk of bias and to synthesize results. We have also revised the conclusion and stated the study limitations. We have now used the PRISMA 2020 abstract checklist in structuring the abstract.

Reviewer 2: In the background, the author mentioned the huge issue of blood pressure control in LMICs, the definition of CHWs. The author stated that the previous reviews focused on NCD but not hypertension, which is not true. There are a few existing review articles that focused on the role of CHWs, training programs, and the task shifting and sharing strategies in BP management. The research gap has not been proposed upon the most recent literature in this field.
o Anand, T. N., Joseph, L. M., Geetha, A. V., Prabhakaran, D., & Jeemon, P. (2019). Task sharing with non-physician health-care workers for management of blood pressure in low-income and middle-income countries: a systematic review and meta-analysis. The Lancet Global Health, 7(6), e761-e771.
o Mills, K. T., Obst, K. M., Shen, W., Molina, S., Zhang, H. J., He, H., ... & He, J. (2018). Comparative effectiveness of implementation strategies for blood pressure control in hypertensive patients: a systematic review and meta analysis. Annals of internal medicine, 168(2), 110-120.
Abdel-All, M., Putica, B., Praveen, D., Abimbola, S., & Joshi, R. (2017). Effectiveness of community health worker training programmes for cardiovascular disease management in low-income and middle-income countries: a systematic review. BMJ open, 7(11), e015529.

Authors: Thank you for this important observation. We have revised the background and included the suggested studies in laying the background of the current study.

Reviewer 2: The objectives of this review are clearly stated with two objectives: what were the main types of CHW-based interventions, and what were the impacts of CHW-led interventions. However, the research findings cannot adequately answer the research question regarding the impacts of CHW-led intervention.

Author: We thank the reviewer for this important comment. We acknowledge that impact of the intervention cannot be answered by our current study findings and have revised the objective 2 to read "what are the outcomes of CHW interventions on the management and control of BP in LMICs?"

Reviewer 2: In the methods section, the author should clearly state how they classified intervention types and outcomes in the data extraction stage and how they synthesized data and information across studies.
Authors: We have now improved the methods section. We have included details of how the interventions were classified and how the qualitative synthesis of the studies was done. We noted that the CHW interventions were characterized by examining the recruitment process for CHW- whether

drawn from government employees or not and the duration of training offered in readiness for the implementation of the interventions. We also categorized the CHW interventions depending on the content and place where the intervention was offered from. The main content of CHW interventions were behavioral communication for lifestyle change and mediator role between patient and health care system. The interventions were either home/community based or clinic based. The studies were also categorized based on whether they had significant reduction in blood pressure which was the main outcome of interest in this review.
Reviewer 2: Search strategy:
o Alternative terms beyond CHW or volunteers, or lay workers can be added such as allied health professionals.
o Use of developing countries or LMICs in the search strategy is problematic as it may miss certain studies even if used as MESH.
o Similarly, only use RCT or N-R CT in MESH to identify study types may miss certain studies.

Authors: We have widened the search strategy to include alternative terminologies related to CHW, LMICs, RCT (see supplementary appendix 1) and repeated the search in two main data bases PubMed and CINAHL. Indeed, the search yielded more articles as shown in figure 1 but after screening we did not find any eligible studies to include in our synthesis. We limited the search to the earlier defined period of January 2000 and September 2020. We belief the search strategy used in this study captured all the relevant studies for the review.

Reviewer 2: The result section has clearly described the characteristics of each study. However, the evidence synthesis has been done on a very superficial level. CHW-involved interventions could include several key types by considering the provider and receptor of the intervention, the dosage and contents. Many of the included studies have multiple components, and CHWs could be considered as both intervention providers and receptors. However, the tables and texts remain descriptive and the key messages learned were not distilled to a higher level.
Authors: We thank the reviewers for this comment. We have improved the synthesis of the evidence on the types of intervention in regard the nature of CHW intervention. We also note that in the text we have a section to address the CHW as recipient of the intervention titled recruitment and training for CHWs. We have highlighted the lack standard training for CHW across the studies and highlighted the length of training and the content of the trainings in the different studies.

Reviewer 2: Similarly, the result section of the outcome and main finding section and Table 1 described the key findings from the trial in a very superficial way. It only described key outcomes in trials, but not the trends in study findings and how the impact of the intervention could be different across studies. We recognize a meta-analysis may not be possible; but synthesizing beyond description of individual studies will be more helpful and expected from the review.

Authors: We acknowledge this comment and agree that bringing out the trend of the study findings and the differences of the outcome across the studies is important to inform future interventions. We have now improved the results section by including a section outcomes trend in relation to intervention implementation strategy across the studies.

Reviewer 2: The discussion is generic to the importance of CHW in the healthcare system and the feasibility and positive impact. There is an increasing number of studies in this field, and the general feasibility and impact have been well-known. The authors haven't discussed how the study findings from this review added to previous reviews or studies in the field and how the study findings could shed light for future intervention design.

Authors: We have improved the discussion by adding the implication of the study findings to future CHW intervention designs.

Reviewer 2: There are some typos/language and formatting issues that need to be improved throughout the manuscripts.
Authors: We have corrected the grammatical and typographical errors

Reviewer 2: The study followed PRISMA guideline 2009. There is PRISMA 2020, and we suggest using the new guideline.

Authors: We thank the reviewer for pointing out this omission on our part. We have now used the new PRISMA guideline of the year 2020.

VERSION 2 – REVIEW

REVIEWER	Perry, HB John Hopkins, International Health
REVIEW RETURNED	23-Dec-2021

GENERAL COMMENTS	There are still numerous minor grammatical mistakes that will need to be corrected by the copy editors.